Prob.6 (Z&T Ch 7, 7.2)

Find the series sol-n to the ↓ Dirichlet pr-m:

Find $u(r,\theta) \in C^2(\Omega) \cap C^0(\bar{\Omega})$ s.t

(1) $$\Delta u(r,\theta) = 0, \quad 0 \le r < 1, \quad -\pi \le \theta < \pi$$

(2) $$u(1,\theta) = f(\theta) \; ; \quad -\pi \le \theta < \pi$$

where $f(\theta)$ is a given $C^0(\partial\Omega)$ f-n (check that ↓ f-ns are $\in C^0(\partial\Omega)$):

(a) $f(\theta) = \theta^2 = \text{poly-1} \in C^0(\partial\Omega)$.

(3) ↗

Note that the conditions met, so the Dirichlet sol-n is

(4) $$u(r,\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n r^n \cos(n\theta) + b_n r^n \sin(n\theta)\right]$$ where

a_0, a_n, b_n's are F-S coef-s of $f(\theta)$:

(5) $$FS\,of\,f = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos(n\theta) + b_n \sin(n\theta)\right]$$ hence

(6) $$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta$$

(7) $$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\,d\theta, \quad n = 1, 2 \ldots$$

(8) $$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\,d\theta, \quad n = 1, 2 \ldots$$

Then
$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}\theta^2\,d\theta = \frac{2}{\pi}\int_0^{\pi}\theta^2\,d\theta = \frac{2}{\pi}\cdot\left(\frac{\theta^3}{3}\right)\Big|_{\theta=0}^{\pi} = \frac{2}{\pi}\cdot\frac{\pi^3}{3} = \frac{2\pi^2}{3}. \Rightarrow$$

(9) $\boxed{a_0 = \frac{2\pi^2}{3}}$

By Pr. 4.(i), (10).

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \theta^2 \cos(n\theta)\, d\theta = \frac{2}{\pi}\int_{0}^{\pi} \theta^2 \cos(n\theta)\, d\theta = \frac{2}{\pi}\cdot\frac{2\pi\cos(n\pi)}{n^2}$$

(10) $\boxed{a_n = \frac{4\cos(n\pi)}{n^2}}$ $\quad n = 1, 2 \ldots$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{\theta^2 \cdot \sin(n\theta)}_{\text{odd}}\, d\theta = 0 \quad \Rightarrow$$

(11) $\boxed{b_n = 0}$ $\quad n = 1, 2 \ldots$ $\quad$ Then, (4) is

(12) $$\boxed{u(r,\theta) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4\cos(n\pi)}{n^2}\cdot r^n \cos(n\theta)}$$

(b) $f(\theta) = |\sin(\theta)| \in C^{\infty}(\Omega) \subseteq C^0(\Omega)$. Hence, $u(r,\theta)$ formula in (4) holds with

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} |\sin(\theta)|\, d\theta = \frac{1}{\pi}\left\{\int_{-\pi}^{0} -\sin(\theta)\, d\theta + \int_{0}^{\pi} \sin\theta\, d\theta\right\} =$$

$$= \frac{1}{\pi}\left\{2\cdot\left[-\cos\theta\right]\Big|_0^{\pi}\right\} = \frac{1}{\pi}(-2)\left[-1-1\right] = \frac{4}{\pi}.$$

(13) $\boxed{a_0 = \frac{4}{\pi}}$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} |\sin(\theta)|\cdot\cos(n\theta)\, d\theta = \ldots = \frac{4}{\pi}\left[\frac{1}{1-4k^2}\right] \quad k = 1, 2, \ldots$$

(14) $\boxed{a_n = \frac{4}{\pi}\cdot\left[\frac{1}{1-4k^2}\right]}$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} |\sin\theta| \sin(n\theta)\, d\theta =$$

$$= \frac{1}{\pi}\left\{ \underbrace{\int_{-\pi}^{0} -\sin(\theta)\cdot\sin(n\theta)\, d\theta}_{=0 \text{ if } n\neq 1} + \underbrace{\int_{0}^{\pi} \sin(\theta)\cdot\sin(n\theta)\, d\theta}_{0 \text{ if } n\neq 1 \text{ by Pr(5)(e) HW6}} \right\}$$

$$= \frac{1}{\pi}\left\{ \int_{-\pi}^{0} -\sin^2\theta\, d\theta + \int_{0}^{\pi} \sin^2(\theta)\, d\theta \right\} = \underbrace{\frac{1}{\pi}\int_{0}^{\pi} \sin^2 d\theta}_{\frac{1}{2}} - \underbrace{\frac{1}{\pi}\int_{-\pi}^{0} \sin^2\theta\, d\theta}_{\frac{1}{2} \text{ by wolf.alfa}} = 0$$

! suspicious might be mistakenly did, but for the time being trusting!

(15) $\boxed{b_n = 0}$ $n = 1, 2, \ldots$ then

Then, by (13) – (15)

(16) $$\boxed{u(r,\theta) = \frac{2}{\pi} + \frac{4}{\pi}\sum_{k=1}^{\infty} \frac{1}{1-4k^2} r^{2k} \cos(2k\theta)}, \quad r \leq 1$$

(c) $f(\theta) = |\theta|(\pi - |\theta|)$.

(17) Again using (4) with

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} |\theta|(\pi - |\theta|)\, d\theta = \frac{\pi^2}{3}$$ by wolfram alpha

(18) $\boxed{a_0 = \frac{\pi^2}{3}}$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} |\theta|(\pi - |\theta|)\cdot\cos(n\theta)\, d\theta = -\frac{2(\pi n - 2\overbrace{\sin(\pi n)}^{=0} + \pi n\cos(\pi n))}{\pi n^3} \quad \forall n$$

$$\Rightarrow a_n = -\frac{2\cancel{\pi} n\left(1+\cos(\pi n)\right)}{\cancel{\pi} n^{\cancel{3}}} = \frac{-2\left(1+\cos(\pi n)\right)}{n^2}.$$

(19) $$\boxed{a_n = -\frac{2\left(1+\cos(\pi n)\right)}{n^2}}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} |\theta|\left(\pi - |\theta|\right)\sin(nx)\,dx = 0.$$

(20) $$\boxed{b_n = 0} \qquad n = 1, 2, ..$$

(21) $$u(r,\theta) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\left[\frac{-2\left(1+\cos(\pi n)\right)}{n^2} r^n \cos(n\theta)\right],$$

if $n = 2k-1$ $k = 1, 2..$ $\Sigma = 0$, so let

$\boxed{n = 2k}$ $k = 1, 2..$

$$u(r,\theta) = \frac{\pi^2}{6} + \sum_{k=1}^{\infty} \frac{-\cancel{2}\cdot\overbrace{\left(1+\cos(2k\pi)\right)}^{1}}{\cancel{2}\,\cancel{4}k^2} r^{2k}\cos(2k\theta) \Rightarrow$$

(22) $$\boxed{u(r,\theta) = \frac{\pi^2}{6} - \sum_{k=1}^{\infty}\frac{1}{k^2}\cdot r^{2k}\cos(2k\theta)}$$